

Trouble shooting tools

Task 1.

Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

```
C:\Users\sketan>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : E3-WDS13.COM

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::d853:f51b:3466:c8a6%6
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:191a:baf1:796a:3c24:d9c4:6b15
    Temporary IPv6 Address. . . . . : 2401:4900:191a:baf1:3c53:a3c5:492e:cf89
    Link-local IPv6 Address . . . . . : fe80::6974:45a0:8874:a935%13
    IPv4 Address. . . . . : 10.183.128.156
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::9409:ff:fe1d:7683%13
                                10.183.128.21
```

Task 2.

Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

```
C:\Users\sketan>ping www.google.com

Pinging www.google.com [2001:4860:482b:7700::] with 32 bytes of data:
Reply from 2001:4860:482b:7700::: time=48ms
Reply from 2001:4860:482b:7700::: time=39ms
Reply from 2001:4860:482b:7700::: time=85ms
Reply from 2001:4860:482b:7700::: time=105ms

Ping statistics for 2001:4860:482b:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 39ms, Maximum = 105ms, Average = 69ms
```

Task 3.

Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

```
C:\Users\sketan>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [2600:140f:8a00::17c1:a55b]
over a maximum of 30 hops:

  0  1  2 ms   3 ms   2 ms  2401:4900:191a:baf1::5e
  1  *      *      *      Request timed out.
  2  47 ms  127 ms  105 ms 2404:a800:2a00:105::d
  3  146 ms 102 ms  101 ms 2404:a800:88::114
  4  115 ms 100 ms  101 ms g2600-140f-8a00-0000-0000-0000-17c1-a55b.deploy.static.akamaitechnologies.com [2600:140f:8a00::17c1:a55b]

Trace complete.
```

Task 4.

Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

```
C:\Users\sketan>nslookup www.spotify.com
Server:      UnKnown
Address:     10.183.128.21

Non-authoritative answer:
Name:        atc.spotify.map.fastly.net
Addresses:   2a04:4e42:31::810
             151.101.211.42
Aliases:     www.spotify.com
```

Task 5.

A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

Hint: Think about checking both network connectivity and DNS resolution.

Command	Purpose
ipconfig	Check whether the computer has a valid IP address and default gateway.
nslookup www.zomato.com	Verify that the DNS server can resolve the website's domain name into an IP address.
ping www.zomato.com	Test whether the website or its server is reachable over the network.

```
C:\Users\sketan>ping www.zoomato.com

Pinging www.zoomato.com [13.248.169.48] with 32 bytes of data:
Reply from 13.248.169.48: bytes=32 time=68ms TTL=247
Reply from 13.248.169.48: bytes=32 time=42ms TTL=247
Reply from 13.248.169.48: bytes=32 time=541ms TTL=247
Reply from 13.248.169.48: bytes=32 time=50ms TTL=247

Ping statistics for 13.248.169.48:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 42ms, Maximum = 541ms, Average = 175ms

C:\Users\sketan>tracert www.zomato.com

Tracing route to e70929.dsca.akamaiedge.net [2600:140f:8a00::17c1:a551]
over a maximum of 30 hops:
```